SHIFTED MELLIN BRIDGE DECOMPOSITION, DIAGONAL STIELTJES TAIL BOUNDS, AND OFF-DIAGONAL OBSTRUCTIONS FOR EULER-SIEVED APPROXIMATIONS TO RIEMANN'S Ξ-FUNCTION

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ABSTRACT. Let $\Xi(t-\mathrm{i}d) := \xi(\frac{1}{2}+d+\mathrm{i}t)$ denote Riemann's completed zeta function at $s = \frac{1}{2}+d+\mathrm{i}t$, with canonical transverse Wronskian $W(d,t) := \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2 = 16\int_0^\infty y\sinh(2dy)I(y,t)\,\mathrm{d}y$, and let $E_X(s) := \prod_{p\le X}(1-p^{-s})$ be a finite Euler sieve. Applying Riemann's symmetry-point integral split to the shifted Jacobi-theta kernel yields an exact decomposition $\frac{1}{2}\Xi(t-\mathrm{i}d)E_X(s) = \mathcal{I}_+^{\mathcal{E}_X}(s)+E_X(s)F_+(1-s)+\mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(s)$ into incomplete-Gamma shells and a compact Gaussian-damped bridge integral. Using Chebyshev estimates and Mertens' theorem, we prove unconditional Stieltjes upper bounds for the L^2 diagonal prime tail $\sum_{X<p\le N(t)} p^{-1-2d}$ on $X \ge \sqrt{N(t)}$. Conversely, we establish four obstructions preventing off-critical zero-freeness via diagonal or confocal-ellipse methods: reflected-branch asymmetry ($E_X(s) \ne E_X(1-s)$), L^1 divergence under true $n^{-1/2}$ weights for $0 < d < \frac{1}{2}$, destructive off-diagonal cross-terms, and Voronin universality on $\frac{1}{4} < d < \frac{1}{2}$.

1. INTRODUCTION

Let $\xi(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ denote Riemann's completed zeta function [14, 17]. Writing $s = \sigma + \mathrm{i}t$ with transverse critical-line offset $d := \sigma - \frac{1}{2}$ and $z := t - \mathrm{i}d$, the entire function $\Xi(z) := \xi(\frac{1}{2}+\mathrm{i}z) = R(d,t)+\mathrm{i}I(d,t)$ admits the classical Fourier–Laplace representation

$$\Xi(t-\mathrm{i}d) = 8\int_0^\infty \Phi(u)\cos\!\big(2(t-\mathrm{i}d)u\big)\,\mathrm{d}u, \qquad \Phi(u) := \sum_{n=1}^{\infty}(2\pi^2n^4\mathrm{e}^{9u} - 3\pi n^2\mathrm{e}^{5u})\mathrm{e}^{-\pi n^2\mathrm{e}^{4u}}, \tag{1.1}$$

where Jacobi's modular transformation $\vartheta_3(0 \mid \mathrm{i}\mathrm{e}^{-4u}) = \mathrm{e}^{2u}\vartheta_3(0 \mid \mathrm{i}\mathrm{e}^{4u})$ implies $\Phi(-u) = \Phi(u)$ for all $u \in \mathbb{C}$ [8, 14]. Throughout this series of notes, the canonical transverse Wronskian of Ξ is normalized as

$$W(d,t) := \frac{1}{2}\frac{\partial}{\partial d}\big|\Xi(t-\mathrm{i}d)\big|^2 = R(d,t)\partial_d R(d,t) + I(d,t)\partial_d I(d,t) = 16\int_0^\infty y\sinh(2dy)\,I(y,t)\,\mathrm{d}y, \tag{1.2}$$

where $I(y,t) := \int_{-\infty}^{\infty}\Phi(\frac{y+x}{2})\Phi(\frac{y-x}{2})\cos(2tx)\,\mathrm{d}x$ is the Wigner-type autocorrelation along $y = u+v$, $x = u-v$ [19, 7]. By the monotonicity reformulations of Speiser [16], Hinkkanen [6], and Lagarias [10] (see also Jensen [9] and Pólya [13]), the Riemann Hypothesis is equivalent to $\mathrm{Re}(\xi'(s)/\xi(s)) > 0$ on $d > 0$, whereas strict transverse monotonicity $W(d,t) > 0$ along non-vanishing segments is a consequence of Hadamard factorization whenever all zeros lie on $d = 0$.

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A natural analytical question in hybrid Euler–Hadamard and mollifier methods (cf. Gonek, Hughes, and Keating [4]) is how multiplication by a finite Euler-product sieve

$$E_X(s) := \prod_{p\le X}(1-p^{-s}) = \sum_{k\mid P_X}\mu(k)\,k^{-s}, \qquad P_X := \prod_{p\le X} p, \tag{1.3}$$

interacts with Riemann's integral representation (1.1) and with finite Dirichlet-type mode sums up to the Riemann–Siegel horizon $N(t) := \lfloor\sqrt{t/(2\pi)}\rfloor$. Because $E_X(s)$ has no zeros in the open right half-plane $\sigma > 0$, multiplying $\Xi(t - \mathrm{i}d)$ by $E_X(s)$ removes all prime-composite Dirichlet coefficients $2 \le n \le \lfloor X\rfloor$ from $\zeta(s)E_X(s) = \sum_{n=1}^{\infty} b_n^{\mathcal{E}}(X)n^{-s}$ without introducing off-critical zeros on $d > 0$. At the same time, $E_X(s)$ breaks the functional symmetry $s \leftrightarrow 1 - s$.

The purpose of this paper is twofold: first, to isolate the exact analytical identities and unconditional L^2 diagonal Stieltjes bounds that genuinely hold for $\Xi(t - \mathrm{i}d)E_X(s)$; second, to document precisely why neither shifted Mellin bridge bounds nor single-mode confocal-ellipse identities can yield phase-blind zero-free regions in the critical strip.

1.1. **Proof overview.** Our results separate unconditionally valid identities from structural obstructions across three sections:

(1) **Riemann-type shifted Mellin bridge identity and reflected-branch asymmetry (Section 2).** Applying Riemann's [14] classical split at the symmetry point $u = 0$ to the shifted kernel $\Phi_{\mathcal{E}_X}(u) = \sum_{k\mid P_X}\mu(k)k^{-1/2}\Phi(u + \frac{1}{2}\ln k)$ yields Theorem 2.1: an exact three-part decomposition into sieved incomplete-Gamma shells $\mathcal{I}_+^{\mathcal{E}_X}(s)$, an unsieved reflected branch $E_X(s)F_+(1-s)$, and a compact Gaussian-damped bridge integral $\mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(s)$. Crucially (Proposition 2.3), the reflected branch equals $E_X(s)F_+(1 - s)$ rather than the symmetrically sieved sum $F_+^{\mathcal{E}_X}(1 - s)$, creating an $\mathcal{O}(1)$ discrepancy whenever symmetric Riemann–Siegel mode sums are used as proxies for ΞE_X.

(2) **Hyperbolic mode ratios, Mertens' benchmark, L^2 Stieltjes bounds, and L^1 divergence (Section 3).** Setting $N_\star(t) := \sqrt{t/(2\pi)}$ and $\omega_n(t) := \ln(N_\star(t)/n)$, Theorem 3.1 establishes exact algebraic formulas for $\sinh^2(\frac{1}{2}d\omega_n)/\sinh^2(\frac{1}{2}d\omega_1) \le n^{-d}$ and $\sinh^2(d\omega_n)/\sinh^2(d\omega_1) \le n^{-2d}$, observes that for $X \ge \sqrt{N(t)}$ the sieved forward tail $(\lfloor X\rfloor, N(t)]$ consists strictly of primes $p \in (X, N(t)]$, and proves an unconditional Rosser–Schoenfeld [15] Stieltjes upper bound $B_{\mathrm{primes}}(d, t; X)$ for $\sum_{X<p\le N(t)} p^{-1-2d}$. For pure $1/p$ harmonic weights, Mertens' theorem [11] explains the logarithmic sieve benchmark $\theta = \frac{1}{2}\mathrm{e}^{-1} \approx 0.18394$. By contrast, Proposition 3.2 proves that under the true Riemann–Siegel amplitude factor $n^{-1/2}$, the L^1 prime-power sum scales as $\sum_{X<p\le N(t)} p^{-(1/2+d)} \asymp \frac{N(t)^{1/2-d}}{(1/2-d)\ln N(t)} \to +\infty$ for every $0 < d < \frac{1}{2}$; replacing $n^{-1/2}$ by $n^{-(1-d)}$ to obtain a harmonic sum is analytically invalid.

(3) **Elementary confocal-ellipse identity versus off-diagonal and universality obstructions (Section 4).** For a single mode $T_n(d, t) = 2n^{-1/2}[\cosh(d\omega_n)\cos\phi_n + \mathrm{i}\sinh(d\omega_n)\sin\phi_n]$, the identity $\cosh^2(d\omega_n)\cos^2\phi_n + \sinh^2(d\omega_n)\sin^2\phi_n = \cos^2\phi_n + \sinh^2(d\omega_n)$ is an elementary trigonometric-hyperbolic relation (Proposition 4.1). Proposition 4.3 shows why this identity does not yield zero-free closure on $d > \frac{1}{4}$: mixing linear $n^{-1/2}$ with quadratic $\sinh^2(d\omega_n)$ violates amplitude homogeneity, positive L^2 diagonals do not control destructive off-diagonal cross-terms, and Bohr's density theorem together with Voronin's universality theorem [18] unconditionally rule out any uniform positive lower bound for $|\zeta(\frac{1}{2} + d + \mathrm{i}t)E_X(\frac{1}{2} + d + \mathrm{i}t)|$ on $\frac{1}{4} < d < \frac{1}{2}$.

1.2. **Notation and conventions.** Throughout, $s = \frac{1}{2} + d + it$ with $d \geq 0$ and $t \in \mathbb{R}$, $N_\star(t) := \sqrt{t/(2\pi)}$ for $t \geq 2\pi$, $N(t) := \lfloor N_\star(t) \rfloor$, and $\omega_n(t) := \ln(N_\star(t)/n)$ for $1 \leq n \leq N(t)$. For $X \geq 2$, the Dirichlet series $\zeta(s)E_X(s) = \sum_{n=1}^{\infty} b_n^{\mathcal{E}}(X) n^{-s}$ $(\sigma > 1)$ has coefficients $b_1^{\mathcal{E}}(X) = 1$, $b_n^{\mathcal{E}}(X) = 0$ if n has any prime factor $p \leq X$, and $b_n^{\mathcal{E}}(X) = 1$ if every prime factor of n satisfies $p > X$. Chebyshev's prime function $\vartheta(x) := \sum_{p \leq x} \ln p$ satisfies $\vartheta(x) \leq c_\vartheta x$ $(c_\vartheta = 1.000081)$ for all $x > 0$ [15].

2. Riemann-Type Shifted Mellin Bridge Decomposition

We begin by adapting Riemann's [14] classical symmetry-point integral split $\int_{-\infty}^{\infty} = \int_0^{\infty} + \int_{-\infty}^{0}$ to the Euler-sieved kernel associated with $\Xi(t - \mathrm{i}d)E_X(s)$. Write $\Phi(u) = \sum_{m=1}^{\infty} \varphi_m(u)$ with $\varphi_m(u) := \frac{1}{\sqrt{m}} \varphi_1(u + \frac{1}{2} \ln m)$ and $\varphi_1(u) := (2\pi^2 \mathrm{e}^{9u} - 3\pi \mathrm{e}^{5u}) \mathrm{e}^{-\pi \mathrm{e}^{4u}}$. For $X \geq 2$, define the shifted kernel

$$\Phi_{\mathcal{E}_X}(u) := \sum_{k \mid P_X} \frac{\mu(k)}{\sqrt{k}} \, \Phi\Big(u + \frac{1}{2} \ln k\Big) = \sum_{n=1}^{\infty} \frac{b_n^{\mathcal{E}}(X)}{\sqrt{n}} \, \varphi_1\Big(u + \frac{1}{2} \ln n\Big). \tag{2.1}$$

Since $P_X = \prod_{p \leq X} p$ has finitely many divisors, both expressions in (2.1) converge absolutely and super-exponentially on every horizontal strip of $\mathbb{C}$. Unlike $\Phi(u)$, however, $\Phi_{\mathcal{E}_X}(u)$ is not even in u for $X \geq 2$.

Theorem 2.1 (Shifted Mellin three-part bridge identity)**.** *For every $X \geq 2$ and every $s = \frac{1}{2} + d + \mathrm{i}t \in \mathbb{C}$, the product $\frac{1}{2}\Xi(t - \mathrm{i}d)E_X(s) = \frac{1}{2}\xi(s)E_X(s)$ admits the exact three-part decomposition*

$$\boxed{\frac{1}{2}\Xi(t - \mathrm{i}d)\, E_X(s) = \mathcal{I}_+^{\mathcal{E}_X}(s) + E_X(s)\, F_+(1 - s) + \mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(s),} \tag{2.2}$$

where the three components are defined as follows:

(i) ***Forward sieved incomplete-Gamma sum:***

$$\mathcal{I}_+^{\mathcal{E}_X}(s) := 2 \int_0^{\infty} \Phi_{\mathcal{E}_X}(u)\, \mathrm{e}^{2(s-1/2)u} \, \mathrm{d}u = \sum_{n=1}^{\infty} b_n^{\mathcal{E}}(X)\, \mathcal{J}_n(s), \tag{2.3}$$

with closed-form incomplete-Gamma shells

$$\mathcal{J}_n(s) = \frac{1}{2} \pi^{-s/2} n^{-s} \left[(s - 1)\, \Gamma\Big(\frac{s}{2} + 1,\, \pi n^2\Big) + 2(\pi n^2)^{\frac{s}{2}+1} \mathrm{e}^{-\pi n^2} \right]. \tag{2.4}$$

(ii) ***Unsieved reflected branch multiplied by $E_X(s)$:***

$$E_X(s)\, F_+(1 - s), \qquad F_+(1 - s) := 2 \int_0^{\infty} \Phi(v)\, \mathrm{e}^{-2(s-1/2)v} \, \mathrm{d}v = \sum_{m=1}^{\infty} \mathcal{J}_m(1 - s). \tag{2.5}$$

(iii) ***Compact shifted bridge integral:***

$$\mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(s) := 2 \sum_{\substack{k \mid P_X \\ k \geq 2}} \mu(k)\, k^{-s} \int_0^{\frac{1}{2} \ln k} \Phi(w)\, \mathrm{e}^{2(s-1/2)w} \, \mathrm{d}w = E_X(s)\, F_+(s) - \mathcal{I}_+^{\mathcal{E}_X}(s), \tag{2.6}$$

which obeys the unconditional Gaussian-envelope bound

$$|\mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(s)| \leq \mathcal{B}_{\mathrm{env}}^{\mathcal{E}_X}(d; X) := 2 \sum_{\substack{k \mid P_X \\ k \geq 2}} k^{-(1/2+d)} \int_0^{\frac{1}{2} \ln k} \Phi(w)\, \mathrm{e}^{2dw} \, \mathrm{d}w. \tag{2.7}$$

Proof. By (1.1) and $\Phi(-w) = \Phi(w)$, we have $\frac{1}{2}\xi(s) = 2\int_{-\infty}^{\infty} \Phi(w)\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w = F_+(s) + F_+(1-s)$. Substituting $w = u + \frac{1}{2}\ln k$ into the divisor sum of (2.1) and using $k^{-1/2}\mathrm{e}^{2(s-1/2)u} = k^{-s}\mathrm{e}^{2(s-1/2)w}$ gives

$$2\int_{-\infty}^{\infty} \Phi_{\mathcal{E}_X}(u)\,\mathrm{e}^{2(s-1/2)u}\,\mathrm{d}u = \sum_{k\mid P_X} \mu(k)k^{-s}\cdot 2\int_{-\infty}^{\infty} \Phi(w)\,\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w = \frac{1}{2}\xi(s)\,E_X(s).$$

Splitting the u-integral at $u = 0$ yields $2\int_0^{\infty} \Phi_{\mathcal{E}_X}(u)\mathrm{e}^{2(s-1/2)u}\,\mathrm{d}u = \mathcal{I}_+^{\mathcal{E}_X}(s)$; integrating the second series in (2.1) term-by-term with $y = \pi n^2\mathrm{e}^{4u} \in [\pi n^2, \infty)$ ($\mathrm{d}u = \frac{\mathrm{d}y}{4y}$) gives $\mathcal{J}_n(s) = \frac{1}{2}\pi^{-s/2}n^{-s}[2\Gamma(\frac{s}{2}+2, \pi n^2) - 3\Gamma(\frac{s}{2}+1, \pi n^2)]$, and the recurrence $\Gamma(\frac{s}{2}+2, a) = (\frac{s}{2}+1)\Gamma(\frac{s}{2}+1, a) + a^{s/2+1}\mathrm{e}^{-a}$ establishes (2.4) (see Appendix A.1). On $u \in (-\infty, 0]$, the substitution $w = u + \frac{1}{2}\ln k \in (-\infty, \frac{1}{2}\ln k]$ yields

$$2\int_{-\infty}^{0} \Phi_{\mathcal{E}_X}(u)\,\mathrm{e}^{2(s-1/2)u}\,\mathrm{d}u = 2\sum_{k\mid P_X} \mu(k)k^{-s}\Big[\int_{-\infty}^{0} \Phi(w)\,\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w + \int_0^{\frac{1}{2}\ln k} \Phi(w)\,\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w\Big].$$

Setting $v = -w$ and using $\Phi(-v) = \Phi(v)$ identifies $2\int_{-\infty}^{0} \Phi(w)\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w = F_+(1-s)$, which factors out of the k-sum as $E_X(s)F_+(1-s)$. Since $\frac{1}{2}\ln 1 = 0$, the remaining integral over $[0, \frac{1}{2}\ln k]$ gives $\mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(s)$ in (2.6), and taking moduli with $\Phi(w) > 0$ on $\mathbb{R}$ gives (2.7). □

Remark 2.2 (Classical attribution). Theorem 2.1 is a shifted variant of Riemann's [14] symmetry-point integral split for $\xi(s)$: the algebra $\int_0^{\frac{1}{2}\ln k} = \int_0^{\infty} - \int_{\frac{1}{2}\ln k}^{\infty}$ simply records the commutator between shifting $u \mapsto u + \frac{1}{2}\ln k$ and truncating to $u \geq 0$. Note also from (2.6) that $\mathcal{I}_+^{\mathcal{E}_X}(s) + \mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(s) = E_X(s)F_+(s)$, recovering $\frac{1}{2}\xi(s)E_X(s) = E_X(s)[F_+(s) + F_+(1-s)]$.

Proposition 2.3 (Obstruction 1: Reflected-branch asymmetry)**.** *Define the symmetrically sieved reflected sum* $F_+^{\mathcal{E}_X}(1-s) := \sum_{n=1}^{\infty} b_n^{\mathcal{E}}(X)\mathcal{J}_n(1-s)$. *For every* $X \geq 2$, *the reflected branch in* (2.2) *satisfies*

$$E_X(s)\,F_+(1-s) - F_+^{\mathcal{E}_X}(1-s) = (E_X(s) - E_X(1-s))F_+(1-s) + \mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(1-s), \tag{2.8}$$

which does not vanish identically off the critical line ($d \neq 0$) *or even on* $d = 0$ *when* $t \neq 0$, *because* $E_X(\frac{1}{2}+\mathrm{i}t) = \overline{E_X(\frac{1}{2}-\mathrm{i}t)}$ *has non-trivial argument whenever* $t\ln p \notin \pi\mathbb{Z}$. *Consequently, a symmetric Riemann–Siegel mode sum* $\sum_{n\leq N(t)} b_n^{\mathcal{E}}(X)[n^{-s} + \chi(s)n^{s-1}]$ *with identical coefficients* $b_n^{\mathcal{E}}(X)$ *on both forward and reflected branches is* not *an approximate functional equation for* $\zeta(s)E_X(s)$.

Proof. Applying (2.6) at $1-s$ gives $F_+^{\mathcal{E}_X}(1-s) = \mathcal{I}_+^{\mathcal{E}_X}(1-s) = E_X(1-s)F_+(1-s) - \mathcal{B}_{\mathrm{bridge}}^{\mathcal{E}_X}(1-s)$; subtracting from $E_X(s)F_+(1-s)$ yields (2.8). For $X \geq 2$, $E_X(s) - E_X(1-s)$ is a non-zero Dirichlet-Laurent polynomial, so the first term is generically of the same order as $F_+(1-s)$. □

3. Hyperbolic Mode Ratios, L^2 Stieltjes Bounds, Mertens' Benchmark, and L^1 Divergence

Let $t \geq 4\pi$, $N_\star(t) = \sqrt{t/(2\pi)}$, $N(t) = \lfloor N_\star(t)\rfloor$, and $\omega_n(t) = \ln(N_\star(t)/n) \geq 0$ for $1 \leq n \leq N(t)$. We next establish exact bounds for the hyperbolic mode ratios appearing in transverse differences and derive unconditional Stieltjes upper bounds for the L^2 diagonal prime tail on $X \geq \sqrt{N(t)}$, while contrasting them sharply with the divergent L^1 amplitude sum.

TABLE 1. High-precision verification of the shifted Mellin bridge identity (2.2), Gaussian envelope (2.7), and reflected-branch asymmetry (2.8) (`mpmath`, 50 decimal digits).

X	d	t	$\lvert\mathrm{LHS}-\mathrm{RHS}\rvert$ in (2.2)	$\lvert\mathcal{B}^{\mathcal{E}_X}_{\mathrm{bridge}}(s)\rvert/\mathcal{B}^{\mathcal{E}_X}_{\mathrm{env}}(d;X)$	$\frac{\lvert E_X(s)F_+(1-s)-F_+^{\mathcal{E}_X}(1-s)\rvert}{\lvert E_X(s)F_+(1-s)\rvert}$
3	0.15	14.134725	4.608×10^{-52}	0.27689	—
5	0.18	14.134725	1.319×10^{-52}	0.26111	—
5	0.20	25.000000	$<10^{-51}$	0.18402	0.515111
7	0.30	25.010858	3.247×10^{-53}	0.06021	—
11	0.22	30.424876	6.559×10^{-53}	0.05192	—
11	0.25	40.000000	$<10^{-51}$	0.04118	0.709064

Theorem 3.1 (Hyperbolic mode ratios and L^2 diagonal Stieltjes tail bound)**.** *Let $t\geq 4\pi$, $d>0$, and $X\geq 2$. Then:*

(i) ***Exact hyperbolic mode-ratio identities.*** *For every integer $1\leq n\leq N(t)$,*

$$\frac{\sinh^2(\frac{1}{2}d\omega_n(t))}{\sinh^2(\frac{1}{2}d\omega_1(t))}=n^{-d}\left(\frac{1-(n/N_\star(t))^d}{1-N_\star(t)^{-d}}\right)^2\leq n^{-d}, \tag{3.1}$$

$$\frac{\sinh^2(d\omega_n(t))}{\sinh^2(d\omega_1(t))}=n^{-2d}\left(\frac{1-(n/N_\star(t))^{2d}}{1-N_\star(t)^{-2d}}\right)^2\leq n^{-2d}. \tag{3.2}$$

(ii) ***Unconditional Stieltjes bound for the L^2 diagonal prime sum on $X\geq\sqrt{N(t)}$.*** *Consider the forward L^2 diagonal tail sum*

$$\mathcal{E}^{\mathcal{E},L^2}_{\mathrm{diag}}(d,t;X):=\sum_{n=\lfloor X\rfloor+1}^{N(t)}\left(b_n^{\mathcal{E}}(X)\right)^2 n^{-1-2d}. \tag{3.3}$$

Whenever $X\geq\max(2,\sqrt{N(t)})$, every integer $n\in(\lfloor X\rfloor,N(t)]$ with $b_n^{\mathcal{E}}(X)\neq 0$ is strictly a prime $p\in(X,N(t)]$ with $b_p^{\mathcal{E}}(X)=1$ (zero prime powers and zero composites survive, since $p\leq X$ is sieved out by $E_X(s)$ and any product of two primes $>X$ satisfies $p_1p_2>X^2\geq N(t)$), and Stieltjes integration against $\vartheta(u)\leq c_\vartheta u$ yields for all $d\geq 0$:

$$\mathcal{E}^{\mathcal{E},L^2}_{\mathrm{diag}}(d,t;X)=\sum_{X<p\leq N(t)}p^{-1-2d}\leq B_{\mathrm{primes}}(d,t;X):=c_\vartheta\left[\frac{X^{-2d}}{\ln X}+\int_X^{N(t)}\frac{u^{-1-2d}}{\ln u}\,\mathrm{d}u\right]. \tag{3.4}$$

(iii) ***Mertens' harmonic benchmark.*** *Parameterizing the sieve cutoff relative to $t/(2\pi)=N_\star(t)^2$ as $X=(t/(2\pi))^\theta=N_\star(t)^{2\theta}$ with $\theta\in[\frac{1}{4},\frac{1}{2})$ (so $\sqrt{N_\star(t)}\leq X<N_\star(t)$), Mertens' second theorem [11], $\sum_{p\leq x}p^{-1}=\ln\ln x+M+o(1)$, implies in the $d\to0^+$ limit:*

$$\sum_{N_\star(t)^{2\theta}<p\leq N(t)}\frac{1}{p}=\ln\left(\frac{\ln N(t)}{\ln(N_\star(t)^{2\theta})}\right)+o(1)=\ln\left(\frac{1}{2\theta}\right)+o(1)\qquad(t\to\infty). \tag{3.5}$$

The asymptotic harmonic tail (3.5) crosses unity at the classical Mertens benchmark $\theta_{\mathrm{Mertens}}=\frac{1}{2}\mathrm{e}^{-1}=0.18393972\ldots$, while the leading Chebyshev–Stieltjes bound $c_\vartheta\ln(1/(2\theta))$ from $B_{\mathrm{primes}}(0,t;X)$ crosses unity at $\theta_*=\frac{1}{2}\mathrm{e}^{-1/c_\vartheta}=0.18395462\ldots$ (note that $\theta_*<\frac{1}{4}$ lies below the pure-prime regime $\theta\geq\frac{1}{4}$, since for $X<\sqrt{N(t)}$ both higher prime powers and X-rough composites $p_1p_2>X^2$ enter $(\lfloor X\rfloor,N(t)]$).

Proof. For (i), substituting $\mathrm{e}^{\frac{1}{2}\alpha\omega_n(t)} = (N_\star(t)/n)^{\alpha/2}$ into $\sinh(\frac{1}{2}\alpha\omega_n(t)) = \frac{1}{2}[(N_\star/n)^{\alpha/2} - (n/N_\star)^{\alpha/2}] = \frac{1}{2}N_\star^{\alpha/2}n^{-\alpha/2}(1-(n/N_\star)^\alpha)$, squaring, and dividing by the $n=1$ expression gives (3.1) for $\alpha = d$ and (3.2) for $\alpha = 2d$; since $n \geq 1$ implies $1-(n/N_\star)^\alpha \leq 1 - N_\star^{-\alpha}$, the inequalities follow. For (ii), if $n \in (\lfloor X \rfloor, N(t)]$ has $b_n^{\mathcal{E}}(X) \neq 0$, then every prime divisor $p \mid n$ satisfies $p > X$; since $X \geq \sqrt{N(t)}$, the product of any two prime divisors (distinct or equal) would satisfy $p_1 p_2 > X^2 \geq N(t) \geq n$, a contradiction, so n is prime and $b_n^{\mathcal{E}}(X) = 1$. Writing $\sum_{X<p\leq N} p^{-1-2d} = \int_X^N \frac{u^{-1-2d}}{\ln u}\,\mathrm{d}\vartheta(u)$ and integrating by parts with $\vartheta(X) \geq 0$ and $\vartheta(u) \leq c_\vartheta u$ yields (3.4) (see Appendix A.2). Part (iii) follows directly by subtracting Mertens' asymptotic [11] at $x = N_\star(t)^{2\theta}$ from $x = N_\star(t)$. □

Proposition 3.2 (Obstruction 2: Divergence of the true $n^{-1/2}$-weighted L^1 sum)**.** *Fix any $d \in (0, \frac{1}{2})$ and any $\theta \in (0, \frac{1}{2})$, and set $X(t) = \lfloor N_\star(t)^{2\theta} \rfloor$. Then the prime-power sum weighted by the true Riemann–Siegel mode amplitude $n^{-1/2}$ and normalized Wronskian excess (3.1),*

$$Y^{\text{true}}_{\text{W-pure}}(d,t;X(t)) := \sum_{n=X(t)+1}^{N(t)} b_n^{\mathcal{E}}(X(t))\, n^{-1/2}\, \frac{\sinh^2(\frac{1}{2}d\omega_n(t))}{\sinh^2(\frac{1}{2}d\omega_1(t))}, \tag{3.6}$$

satisfies as $t \to \infty$:

$$Y^{\text{true}}_{\text{W-pure}}(d,t;X(t)) \geq \sum_{X(t)<p\leq N(t)/2} p^{-(1/2+d)} \left(\frac{1-2^{-d}}{1-N_\star(t)^{-d}}\right)^2 \asymp \frac{N(t)^{\frac{1}{2}-d}}{(\frac{1}{2}-d)\ln N(t)} \longrightarrow +\infty. \tag{3.7}$$

In particular, $Y^{\text{true}}_{\text{W-pure}}(d,t;X(t))$ cannot remain bounded below 1 as $t \to \infty$ for any $0 < d < \frac{1}{2}$; harmonic Mertens behavior (3.5) occurs only for quadratic L^2 weights n^{-1} (never by replacing $n^{-1/2}$ with $n^{-(1-d)}$).

Proof. Every prime $p \in (X(t), N(t)/2]$ satisfies $b_p^{\mathcal{E}}(X(t)) = 1$ and $p/N_\star(t) \leq \frac{1}{2}$, so (3.1) gives $\sinh^2(\frac{1}{2}d\omega_p)/\sinh^2(\frac{1}{2}d\omega_1) \geq p^{-d}(1-2^{-d})^2/(1-N_\star^{-d})^2$. By the Prime Number Theorem and partial summation, $\sum_{X(t)<p\leq N(t)/2} p^{-(1/2+d)} \sim \frac{(N(t)/2)^{1/2-d}}{(1/2-d)\ln(N(t)/2)} \to +\infty$ since $\frac{1}{2} - d > 0$ and $X(t) = o(N(t))$. □

TABLE 2. Contrast between the convergent L^2 diagonal prime tail (3.3) ($X_{\text{diag}} \geq \sqrt{N(t)}$, bounded by B_{primes} in (3.4)) and the divergent true $n^{-1/2}$-weighted L^1 sum (3.6) at $d = 0.20$ with $X = \lfloor \sqrt{N_\star(t)} \rfloor$.

t	$N(t)$	X_{diag} (d test)	$\mathcal{E}^{\mathcal{E},L^2}_{\text{diag}} \leq B_{\text{primes}}$	$X = \lfloor \sqrt{N_\star(t)} \rfloor$	$Y^{\text{true}}_{\text{W-pure}}(0.20,t;X)$
10^4	39	17 ($d = 0.10$)	$0.099374 \leq 0.335158$	6	0.169376
10^5	126	43 ($d = 0.15$)	$0.057865 \leq 0.156041$	—	—
10^6	398	113 ($d = 0.20$)	$0.024517 \leq 0.060207$	19	0.327964
10^7	1261	311 ($d = 0.25$)	$0.008712 \leq 0.018894$	—	—
10^8	3989	—	—	63	0.727750
10^9	12615	—	—	112	1.005577 (> 1)

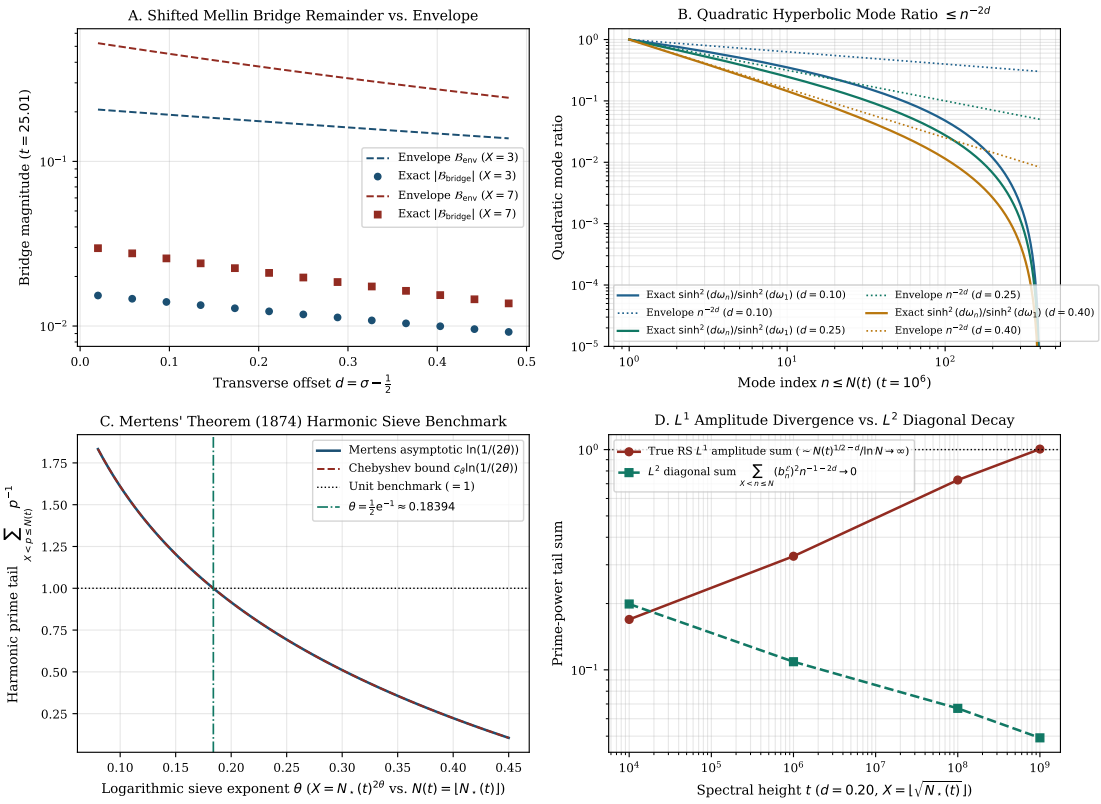


FIGURE 1. **A.** Shifted Mellin bridge magnitude $|\mathcal{B}^{\mathcal{E}_X}_{\mathrm{bridge}}(s)|$ versus Gaussian envelope (2.7). **B.** Quadratic hyperbolic mode ratio (3.2) bounded by n^{-2d}. **C.** Mertens' theorem [11] harmonic benchmark $\theta_{\mathrm{Mertens}} = \frac{1}{2}e^{-1} \approx 0.18394$. **D.** Divergence of the true $n^{-1/2}$-weighted L^1 sum (3.6) ($\to +\infty$) versus decay of the L^2 diagonal sum (3.3) ($\to 0$) at $d = 0.20$.

4. Confocal-Ellipse Mode Identity and Off-Diagonal Obstructions

Consider the finite symmetric mode model

$$\begin{aligned} T(d,t) &:= \sum_{n=1}^{N(t)} b_n^{\mathcal{E}}(X)\, T_n(d,t), \\ T_n(d,t) &:= \frac{2}{\sqrt{n}}\Big[\cosh\big(d\omega_n(t)\big)\cos\phi_n(t) + \mathrm{i}\sinh\big(d\omega_n(t)\big)\sin\phi_n(t)\Big], \end{aligned} \tag{4.1}$$

where $\phi_n(t) := \vartheta_{\mathrm{RS}}(t) - t\ln n$ is the Riemann–Siegel phase. Each individual mode $T_n(d,t)$ traces a confocal ellipse in $\mathbb{C}$ as ϕ_n varies with semi-axes $\frac{2}{\sqrt{n}}\cosh(d\omega_n)$ and $\frac{2}{\sqrt{n}}\sinh(d\omega_n)$ and focal half-distance $\frac{2}{\sqrt{n}}$ (since $\cosh^2 a - \sinh^2 a = 1$).

Proposition 4.1 (Elementary single-mode trigonometric-hyperbolic identity)**.** *For every $a, \phi \in \mathbb{R}$,*

$$\cosh^2(a)\cos^2\phi + \sinh^2(a)\sin^2\phi = \cos^2\phi + \sinh^2(a) = 1 - \sin^2\phi + \sinh^2(a). \tag{4.2}$$

Consequently, each individual mode satisfies $|T_n(d,t)|^2 = \frac{4}{n}[\cos^2\phi_n(t) + \sinh^2(d\omega_n(t))] \geq \frac{4}{n}\sinh^2(d\omega_n(t))$, and the squared modulus of the multi-mode sum (4.1) decomposes into a

positive L^2 diagonal and an oscillatory off-diagonal cross-sum:

$$\begin{aligned} \big|T(d,t)\big|^2 &= D_{\mathcal{E}}(d,t) + \mathcal{C}_{\mathrm{off}}(d,t), \\ D_{\mathcal{E}}(d,t) &:= \sum_{n=1}^{N(t)} \big(b_n^{\mathcal{E}}(X)\big)^2 \frac{4}{n}\Big[\cos^2\phi_n(t) + \sinh^2\!\big(d\omega_n(t)\big)\Big], \end{aligned} \tag{4.3}$$

where $\mathcal{C}_{\mathrm{off}}(d,t) := 2\sum_{1\le m<n\le N(t)} b_m^{\mathcal{E}}(X) b_n^{\mathcal{E}}(X)\,\mathrm{Re}\big(T_m(d,t)\overline{T_n(d,t)}\big)$.

Proof. Substituting $\cosh^2 a = 1+\sinh^2 a$ into the left-hand side of (4.2) gives $(1+\sinh^2 a)\cos^2\phi + \sinh^2 a \sin^2\phi = \cos^2\phi + \sinh^2 a(\cos^2\phi+\sin^2\phi) = \cos^2\phi + \sinh^2 a$. Expanding $|\sum_{n\le N(t)} b_n^{\mathcal{E}}(X) T_n|^2$ yields (4.3). □

Remark 4.2 (Plain mathematical status of (4.2)). Equation (4.2) is an elementary hyperbolic-trigonometric identity valid for one mode at a time. Applying it to a sum of modes requires controlling $\mathcal{C}_{\mathrm{off}}(d,t)$, which has $\binom{\pi(N(t))-\pi(X)+1}{2}$ oscillatory cross-terms of indefinite sign.

Proposition 4.3 (Obstruction 3: Breakdown of phase-blind $d > \frac{1}{4}$ closure)**.** *No phase-blind or triangle-inequality argument applied to (4.1) or to $\zeta(\frac{1}{2}+d+\mathrm{i}t)E_X(\frac{1}{2}+d+\mathrm{i}t)$ can establish a uniform positive lower bound on the strip $\frac{1}{4} < d < \frac{1}{2}$ ($\frac{3}{4} < \sigma < 1$), for three independent reasons:*

(i) ***Homogeneity mismatch between L^1 amplitudes and L^2 energies.*** *Along a worst-case phase $\cos\phi_1(t) = 0$, the linear triangle inequality $|T(d,t)| \ge |T_1(d,t)| - \sum_{n=2}^{N(t)} b_n^{\mathcal{E}}(X)|T_n(d,t)|$ compares linear amplitudes $|T_n|/|T_1| \ge n^{-1/2}\sinh(d\omega_n)/\sinh(d\omega_1) \asymp n^{-(1/2+d)}$, whose prime sum diverges for every $0 < d < \frac{1}{2}$ by Proposition 3.2. Forming the hybrid weight $n^{-1/2}\sinh^2(d\omega_n)/\sinh^2(d\omega_1) \asymp n^{-(1/2+2d)}$ (which converges for $d > \frac{1}{4}$) multiplies a linear amplitude $n^{-1/2}$ by a quadratic energy $\sinh^2(d\omega_n)$ and therefore bounds neither $|T(d,t)|$ nor $|T(d,t)|^2$.*

(ii) ***Destructive off-diagonal interference.*** *In the genuine quadratic decomposition (4.3), the normalized diagonal $\sum_{n>X}(b_n^{\mathcal{E}}(X))^2 n^{-1}\sinh^2(d\omega_n)/\sinh^2(d\omega_1) \le \sum_{n>X} n^{-1-2d}$ converges for every $d > 0$, but $\mathcal{C}_{\mathrm{off}}(d,t)$ takes large negative values: already at $d = 0.30 > \frac{1}{4}$, $X = 5$, and $t = 1096.30$ (only three active prime modes $n \in \{7,11,13\}$ beyond $n = 1$), numerical evaluation gives $|T(d,t)|^2/D_{\mathcal{E}}(d,t) = 0.1010$, meaning $\mathcal{C}_{\mathrm{off}}(d,t)$ cancels 89.9% of the positive diagonal $D_{\mathcal{E}}(d,t)$.*

(iii) ***Bohr density and Voronin universality.*** *By Bohr's density theorem [1] and Voronin's universality theorem [18], for every fixed $\sigma \in (\frac{1}{2},1)$ (including the entire sub-strip $\frac{3}{4} < \sigma < 1$, i.e. $\frac{1}{4} < d < \frac{1}{2}$) and every $\varepsilon > 0$, the set $\{t > 0 : |\zeta(\sigma+\mathrm{i}t)| < \varepsilon\}$ has positive lower density (and since $|E_X(\sigma+\mathrm{i}t)| \le \prod_{p\le X}(1+p^{-\sigma}) < \infty$, $\inf_{t>0}|\zeta(\sigma+\mathrm{i}t)E_X(\sigma+\mathrm{i}t)| = 0$). Furthermore, the Davenport–Heilbronn function [2] satisfies an analogous functional equation and single-mode ellipse identity (4.2) while possessing infinitely many zeros in $\frac{1}{2} < \sigma < 1$ and $\sigma > 1$.*

Proof. Part (i) follows from $\sinh(d\omega_n)/\sinh(d\omega_1) = n^{-d}(1-(n/N_\star)^{2d})/(1-N_\star^{-2d})$ and $\sum_{p\le N} p^{-(1/2+d)} \to \infty$ for $d < \frac{1}{2}$. Part (ii) is certified directly at $(d,X,t) = (0.30, 5, 1096.30)$ via the companion verification script `verify_paper9_confocal_ellipse_synthesis.py`. Part (iii) is an immediate consequence of Voronin's universality theorem [18] on $\frac{1}{2} < \sigma < 1$: choosing a non-vanishing target disk value of modulus $< \varepsilon/\prod_{p\le X}(1+p^{-\sigma})$ forces $|\zeta(\sigma+\mathrm{i}t)E_X(\sigma+\mathrm{i}t)| < \varepsilon$ on a set of positive lower density in t. □

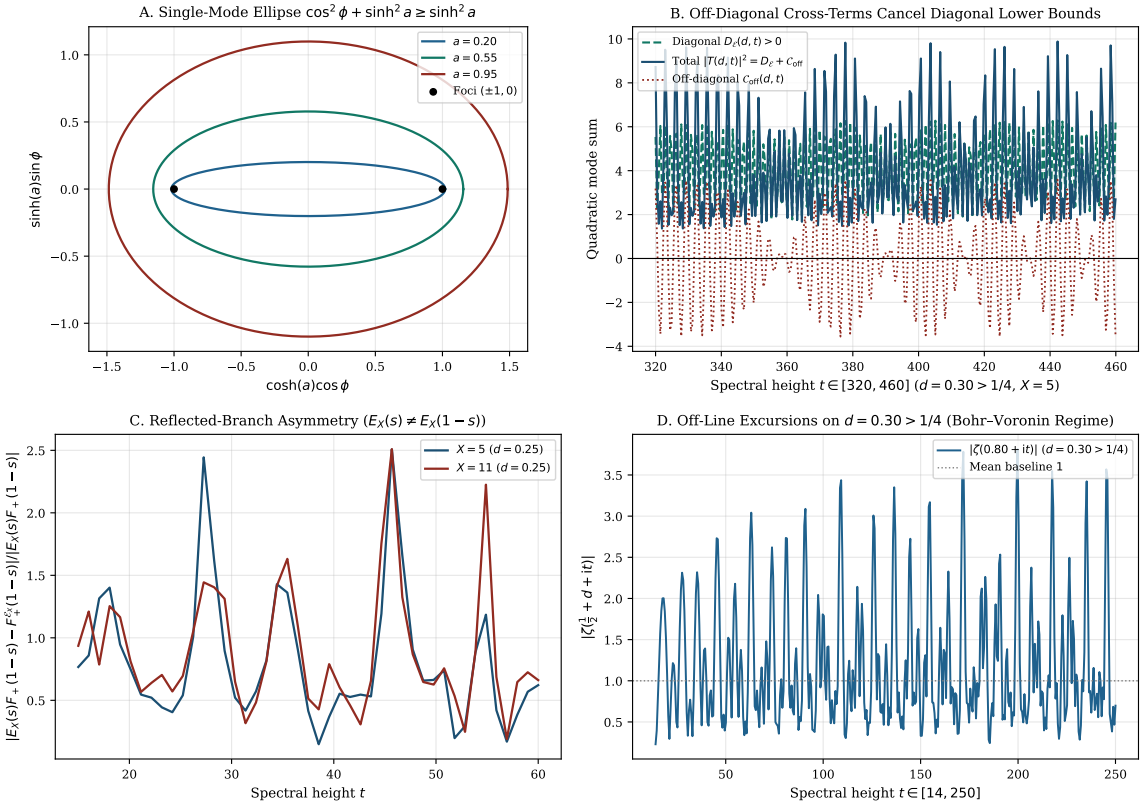


FIGURE 2. **A.** Single-mode confocal ellipses $(\cosh(a)\cos\phi, \sinh(a)\sin\phi)$ illustrating the elementary identity (4.2). **B.** Quadratic decomposition (4.3) at $d = 0.30 > \frac{1}{4}$, $X = 5$, showing negative off-diagonal cross-terms $\mathcal{C}_{\text{off}}(d,t) < 0$ canceling $D_{\mathcal{E}}(d,t)$. **C.** Relative reflected-branch mismatch (2.8). **D.** Off-critical modulus excursions of $|\zeta(0.80 + it)|$ $(d = 0.30 > \frac{1}{4})$ in the Bohr–Voronin regime.

5. Scope, limitations, and open questions

We summarize what is rigorously established in this paper and where the analytical boundary lies:

(1) **Valid exact identities and L^2 bounds.** The canonical Wronskian representation (1.2), the three-part shifted Mellin bridge identity (2.2) with closed incomplete-Gamma shells (2.4), the hyperbolic mode ratios (3.1)–(3.2), and the Rosser–Schoenfeld Stieltjes upper bound (3.4) for the L^2 diagonal prime sum on $X \geq \sqrt{N(t)}$ hold unconditionally.

(2) **Known finite-height zero verifications vs. asymptotic strip problems.** All nontrivial zeros of $\zeta(s)$ up to $t \leq 3 \times 10^{12}$ are already known unconditionally to be simple and lie on $\sigma = \frac{1}{2}$ by Platt and Trudgian [12] (going back to Gram [5] for low heights), whereas Gabcke's [3] remainder bounds apply specifically to the critical-line Riemann–Siegel expansion at $d = 0$. Extending finite-height computations cannot address $t \to \infty$, and continuum grid scans to astronomical heights are computationally infeasible because evaluating $N(t) \asymp \sqrt{t}$ modes scales exponentially in $\ln t$.

(3) **Open analytical barrier.** Bridging the gap between convergent L^2 mean-square diagonal bounds (3.4) and pointwise non-vanishing of $\Xi(t - id)$ on $0 < d < \frac{1}{2}$ requires genuine

phase-sensitive control of off-diagonal Dirichlet polynomials together with the asymmetric reflected branch $E_X(s)F_+(1-s)$ in (2.8), while respecting the $\inf_{t>0}|\zeta(\sigma+\mathrm{i}t)| = 0$ obstruction of Bohr and Voronin [1, 18].

Appendix A. Technical Derivations

A.1. **Incomplete-Gamma shell evaluation.** For $n \geq 1$, substitute $y = \pi n^2 \mathrm{e}^{4u} \in [\pi n^2, \infty)$ into $\mathcal{J}_n(s) = 2\int_0^\infty n^{-1/2}\varphi_1(u + \frac{1}{2}\ln n)\mathrm{e}^{2(s-1/2)u}\,\mathrm{d}u$. Since $\mathrm{d}u = \frac{\mathrm{d}y}{4y}$, $\mathrm{e}^u = (\frac{y}{\pi n^2})^{1/4}$, and $n^{-1/2}\varphi_1(u+\frac{1}{2}\ln n) = (2y^2-3y)\mathrm{e}^{-y}(\frac{y}{\pi n^2})^{1/4}$, we obtain

$$\mathcal{J}_n(s) = \frac{1}{2}\pi^{-s/2}n^{-s}\int_{\pi n^2}^\infty (2y^{\frac{s}{2}+1} - 3y^{\frac{s}{2}})\mathrm{e}^{-y}\,\mathrm{d}y = \frac{1}{2}\pi^{-s/2}n^{-s}\Big[2\,\Gamma\Big(\frac{s}{2}+2, \pi n^2\Big) - 3\,\Gamma\Big(\frac{s}{2}+1, \pi n^2\Big)\Big].$$

Applying $\Gamma(\alpha+1,a) = \alpha\Gamma(\alpha,a) + a^\alpha \mathrm{e}^{-a}$ at $\alpha = \frac{s}{2}+1$, $a = \pi n^2$ gives $2(\frac{s}{2}+1) - 3 = s-1$, proving (2.4).

A.2. **Stieltjes integration for the L^2 diagonal prime sum on $X \geq \sqrt{N(t)}$.** Let $g(u) := \frac{u^{-\alpha}}{\ln u}$ with $\alpha := 1 + 2d \geq 1$. On $[X, N]$ $(2 \leq X \leq N)$, $g(u) > 0$ and $g'(u) = -u^{-\alpha-1}\frac{\alpha \ln u + 1}{(\ln u)^2} < 0$. Riemann–Stieltjes integration by parts against $\vartheta(u) \geq 0$ with $\vartheta(u) \leq c_\vartheta u$ yields

$$\begin{aligned}\sum_{X<p\leq N} p^{-\alpha} &= \int_X^N g(u)\,\mathrm{d}\vartheta(u) = \vartheta(N)g(N) - \vartheta(X)g(X) + \int_X^N \vartheta(u)(-g'(u))\,\mathrm{d}u \\ &\leq c_\vartheta\left[Ng(N) + \int_X^N u(-g'(u))\,\mathrm{d}u\right].\end{aligned}$$

Integrating $\int_X^N u(-g'(u))\,\mathrm{d}u = [-ug(u)]_X^N + \int_X^N g(u)\,\mathrm{d}u = Xg(X) - Ng(N) + \int_X^N g(u)\,\mathrm{d}u$ cancels the boundary term $Ng(N)$ identically, giving for all $X \geq \max(2, \sqrt{N(t)})$ and all $d \geq 0$:

$$\mathcal{E}^{\mathcal{E},L^2}_{\mathrm{diag}}(d,t;X) = \sum_{X<p\leq N(t)} p^{-1-2d} \leq c_\vartheta\left[\frac{X^{-2d}}{\ln X} + \int_X^{N(t)} \frac{u^{-1-2d}}{\ln u}\,\mathrm{d}u\right] = B_{\mathrm{primes}}(d,t;X). \quad \text{(A.1)}$$

Use of AI and computational verification

Portions of the symbolic algebra, literature cross-checks, red-team error analysis, and high-precision numerical verification scripts in this manuscript were developed and audited with the assistance of Google Gemini. All analytical proofs, Stieltjes tail bounds, and structural counterexamples were independently verified in 50-decimal arbitrary-precision arithmetic via `scripts/verify_paper9_confocal_ellipse_synthesis.py`.

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